

Kieran Douglas

kieran@tilquhillie.com

(520)576-0707

LinkedIn

GitHub

Website

EDUCATION

University of California, Davis

2025–2027

M.S. in Agricultural and Resource Economics

Davis, CA

- Emphasis: Development Economics — Thesis: *Immigration Enforcement, Labor Scarcity, and Mechanization in U.S. Agriculture*
- Selected coursework: Real Analysis, Linear Algebra, Multivariable Calculus, Advanced Econometrics, Advanced Microeconomics.

University of Arizona

2021–2025

*B.A. in Economics, summa cum laude; Minor in Population Health
Data Science*

Tucson, AZ

RESEARCH EXPERIENCE

University of California, Davis

*Graduate Research Assistant | Agricultural and Resource Economics | Apr. 2026–present
Immigration Enforcement, Labor Scarcity, and Mechanization in U.S. Agriculture*

- Examine whether exposure to Secure Communities–induced deportation shocks alters agricultural input choices in U.S. counties.
- Construct and analyze a county-level panel linking immigration-enforcement measures to agricultural outcomes; estimate effects of enforcement exposure on labor expenditures and labor-saving technology adoption.
- Assess heterogeneous effects by counties' baseline vulnerability to enforcement exposure and test mechanisms consistent with uncertainty-induced labor replacement.
- Supported by a \$35,000 Giannini Foundation Mini-Grant and a fellowship with the Transactional Records Access Clearinghouse (TRAC), 2026–27.

Applied International Development Economics Lab (AIDE), University of Arizona

Student Researcher | Aug. 2023–present

Mining Meaning: Measurement Error in Literature Reviews Using Generative AI

- Co-author a paper that evaluates measurement error in LLM-assisted extraction of instrumental-variable choices and study characteristics from economics papers using weather-based instruments.
- Develop reproducible workflows for processing large-scale academic corpora and extracting structured metadata across model families and prompting strategies.

Variable Selection in Economic Applications of Remotely Sensed Weather Data: Evidence from the LSMS

- Review 174 studies to document common rainfall metrics used as instruments and quantify the prevalence of ad hoc variable selection across outcomes.
- Contribute code and prepare World Bank Living Standards Measurement Study (LSMS) data for empirical analysis.

Integrating Weather and Land Cover Data into Geospatial Impact Evaluations

- Contribute literature review and synthesis for a book chapter on the appropriate use of weather, vegetation, land-cover, and extreme-event data in geospatial impact evaluation.

Independent and Pro Bono Research

Researcher | Jan. 2025–present

Civicism

- Develop a civic-engagement platform that maps users, elected officials, and ballot initiatives into a shared ideological space.
- Build an LLM-based schema that extracts policy-relevant metadata from public records and local ballot materials.
- Construct a test bed for evaluating whether embeddings of heterogeneous textual metadata can proxy for ideological preferences.

Pro Bono Research Volunteer | Digital Ground Game | May 2026–present

- Contribute to survey development, data collection, and analysis for a longitudinal study of web-based political action.
- Develop state- and local-government tracking products that make political information more accessible.

PUBLICATIONS AND PRESENTATIONS

Working Papers

- **K. Douglas**, J. D. Michler, and A. Josephson. “Mining Meaning: Measurement Error in Literature Reviews Using Generative AI.”
- Agme, C. D., **K. Douglas**, J. D. Michler, and A. Josephson. “Variable Selection in Economic Applications of Remotely Sensed Weather Data: Evidence from the LSMS.”
- **K. Douglas**, A. Shenoy, S. Boucher, and T. Pearson. “Immigration Enforcement, Labor Scarcity, and Mechanization in U.S. Agriculture.” Master’s thesis in progress, University of California, Davis.

Selected Presentations

- “Immigration Enforcement, Labor Scarcity, and Mechanization in U.S. Agriculture.” BITSS RT2 Workshop, University of California, Berkeley, May 2026.
- “Mining Meaning: Conducting AI-Assisted Reviews of Economic Literature.” Berkeley Initiative for Transparency in the Social Sciences Annual Meeting, University of California, Berkeley, Apr. 2026.
- “Mining Meaning: Conducting AI-Assisted Reviews of Economic Literature.” VIP and CURES Poster and Presentation Session, University of Arizona, Apr. 2025.
- “Ambient Air Pollution Concentrations and Poverty in American Cities.” Equitable Agriculture and Environmental Management Program, University of California, Santa Barbara, Aug. 2024.
- “Exploring Complications With Instrumental Variable Selection and Their Solutions.” Undergraduate Research Opportunities Consortium Summer Colloquium and Poster Session, University of Arizona, Aug. 2024.

Publications

- *Exploring Complications With Instrumental Variable Selection and Their Solutions.* (2024). Undergraduate Research Opportunities Consortium.

HONORS, AWARDS, AND FUNDING

Giannini Foundation Mini-Grant (\$35,000)	2026–27
BITSS RT2 Program Funding, University of California, Berkeley	2026
Non-Residential Supplemental Tuition Fellowship, University of California, Davis	2025–26

First Place, Choice Awards, UA Vertically Integrated Projects Session	2025
Undergraduate Research Opportunities Consortium (UROC) Summer Research Institute	2024
Wildcat Recognition Tuition Award, University of Arizona	2021–25
Dean’s List with Distinction, University of Arizona	2021–25

PROFESSIONAL EXPERIENCE AND AFFILIATIONS

University of California, Davis

Teaching Assistant / Reader | Agricultural and Resource Economics | Aug. 2025–present

- Support undergraduate economics instructors through grading, exam proctoring, and student assistance.

Private Tutoring

Tutor | Self-employed | Aug. 2025–present

- Provide one-on-one instruction in AP and undergraduate microeconomics.
- Develop individualized study plans and supplemental instructional materials.

Transactional Records Access Clearinghouse (TRAC)

Research Fellow | Jun. 2026–present

- Use immigration-enforcement data to study uncertainty-induced labor replacement in U.S. agriculture.

TECHNICAL SKILLS

R, Git/GitHub, L^AT_EX, reproducible workflows, replication-package development, Python, Agentic workflows, semantic modeling, GIS, Prompt Engineering, Machine Learning, survey design.